What are Automations ?
Answer:

Automation is the process of using technology to perform tasks without human intervention, making repetitive or time-consuming activities faster and more efficient.

Automations:

- Follow a fixed, rule-based process.
- Great for repetitive, predictable tasks.
- Example: An email sent to every new customer as soon as they sign up.

Automation are Deterministic and Predefined:

Deterministic: It means not fuzzy in nature which means they don't change, they can be defined by rule and,

Predefined: It can be pre-defined like if it happens then it should be done.

Example:

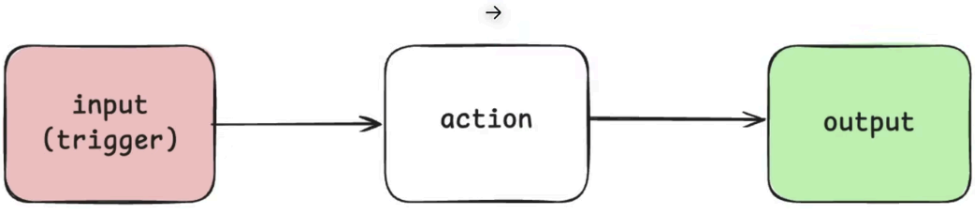
- Automatically sending a welcome email to a new subscriber after they sign up for a newsletter.
- A machine that makes coffee at 7:00 AM every day.

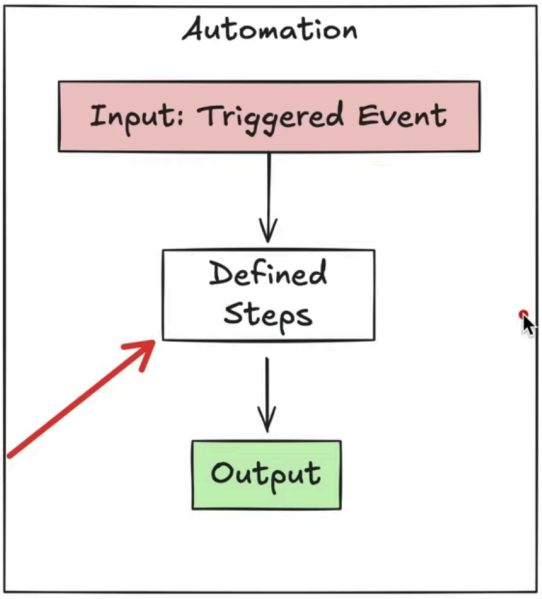
What are Agents ?
Answer:

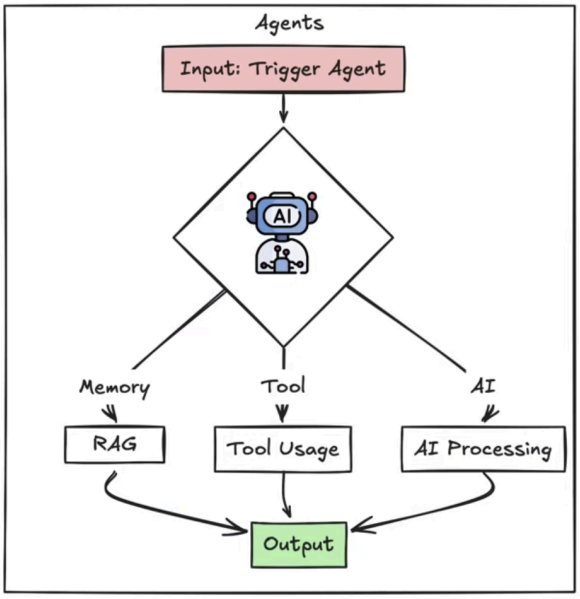
- An AI agent is a node that uses artificial intelligence to perform smart tasks, like analyzing text or making decisions.

- An AI agent is like a super-smart assistant. It's the one who read the room (or data) and figures out what to do next.

Flow







What is an n8n ?
Answer:

- N8N is an open-source, no-code tool that lets you connect apps and services to automate workflows and integrate AI-driven tasks.

- We can start from free and it can also be hosted locally or on cloud.

- It has more than 500+ integrations with common applications and AI.

- Since this is a no-code tool, we can easily use and master it to build our automation and agents.

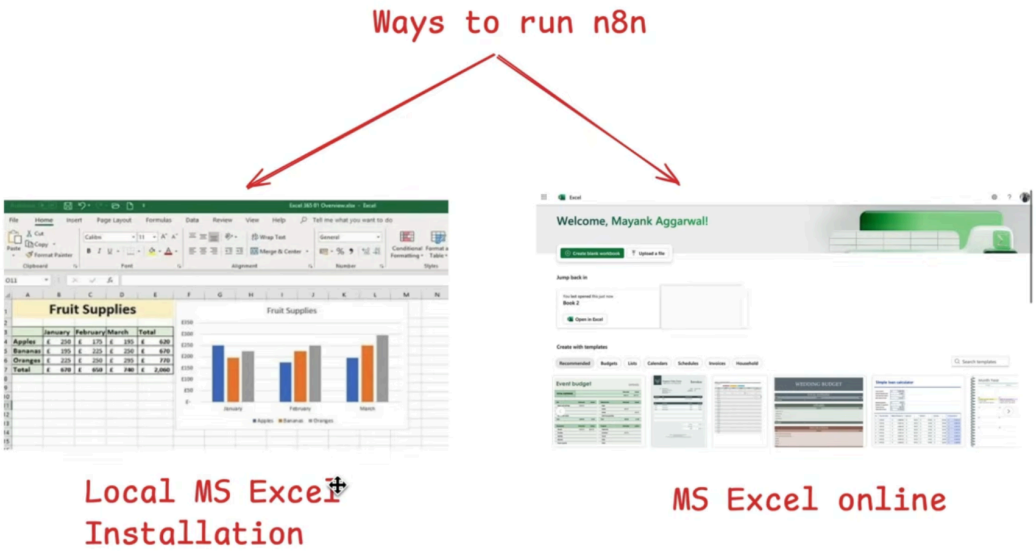
- It's free to start, beginner-friendly, and packed with integrations. No coding required - just drag, drop, and connect.

- There's a huge N8N community with templates and tutorials to guide us.

- URL for documentation: https://n8n.io/itops/
- Shortcut keyword: https://docs.n8n.io/keyboard-shortcuts/

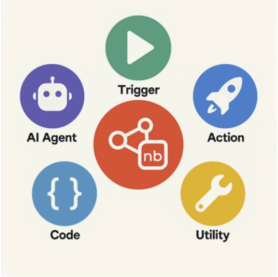
Ways to run n8n:

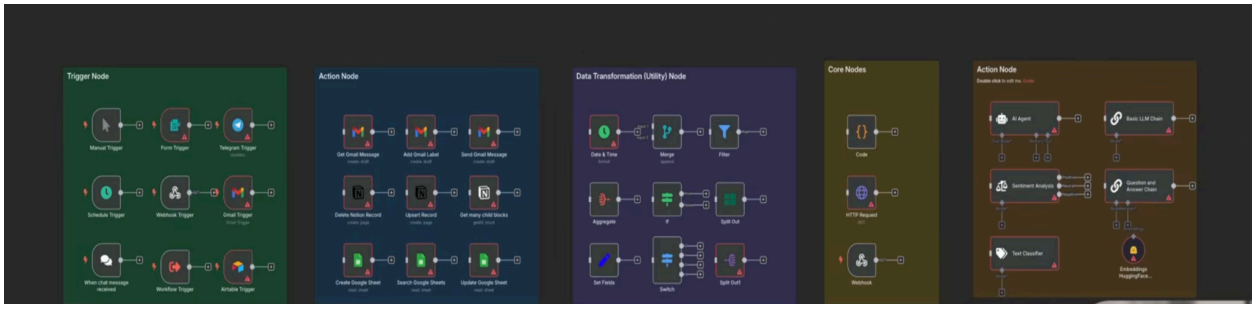
1.) Directly on n8n. (free trial 14 days) + Monthly Pricing.
2.) Local.
3.) Self hosted on a server.



Nodes in n8n:

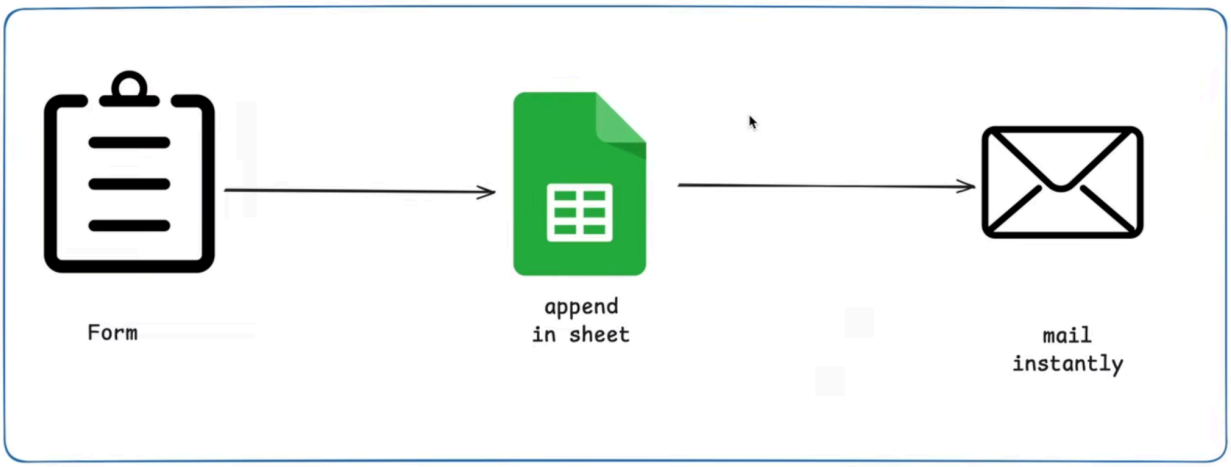
A node is a single building block in an N8N workflow, each performing a specific job like triggering an event or processing data.

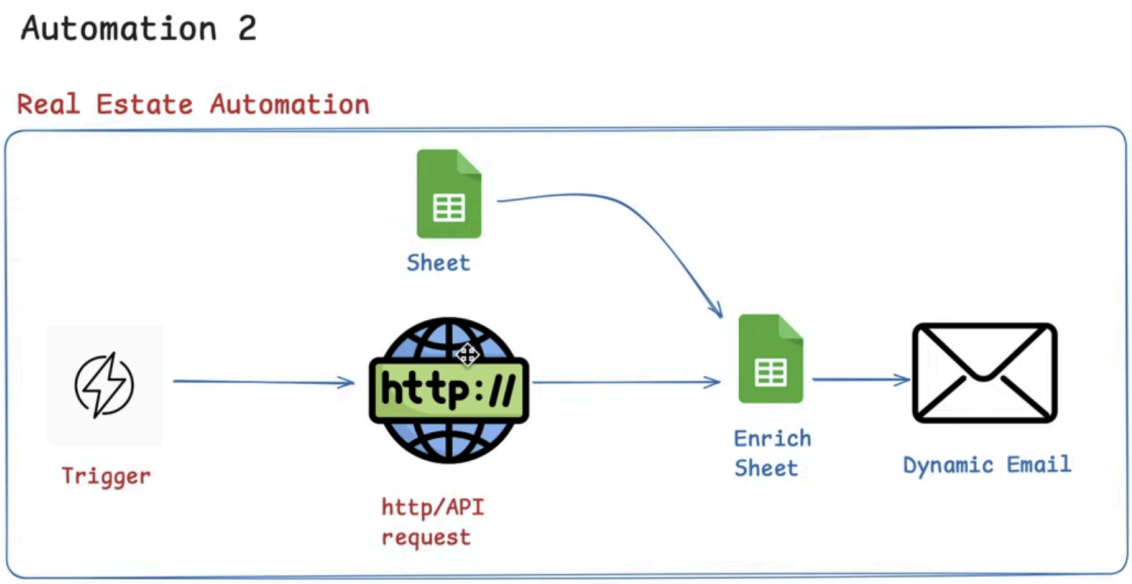




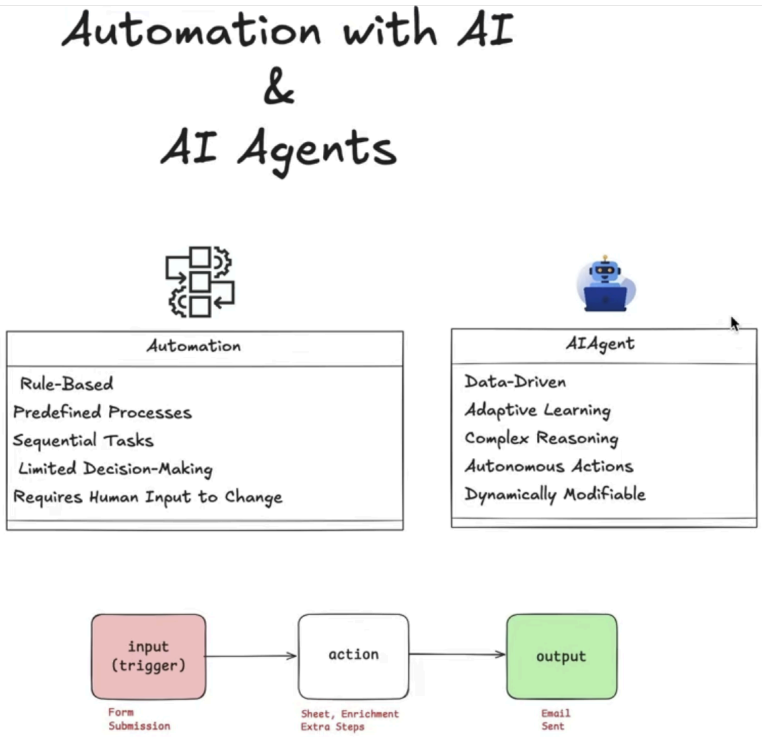
Automation 1

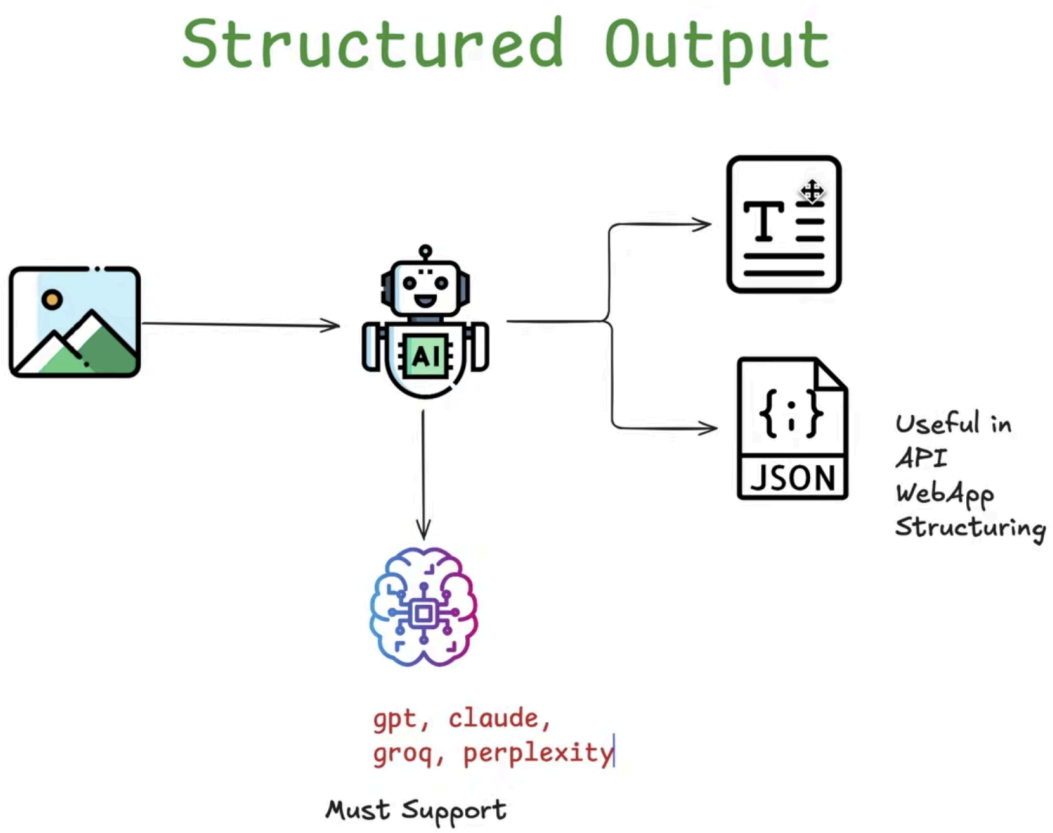
Form Automation





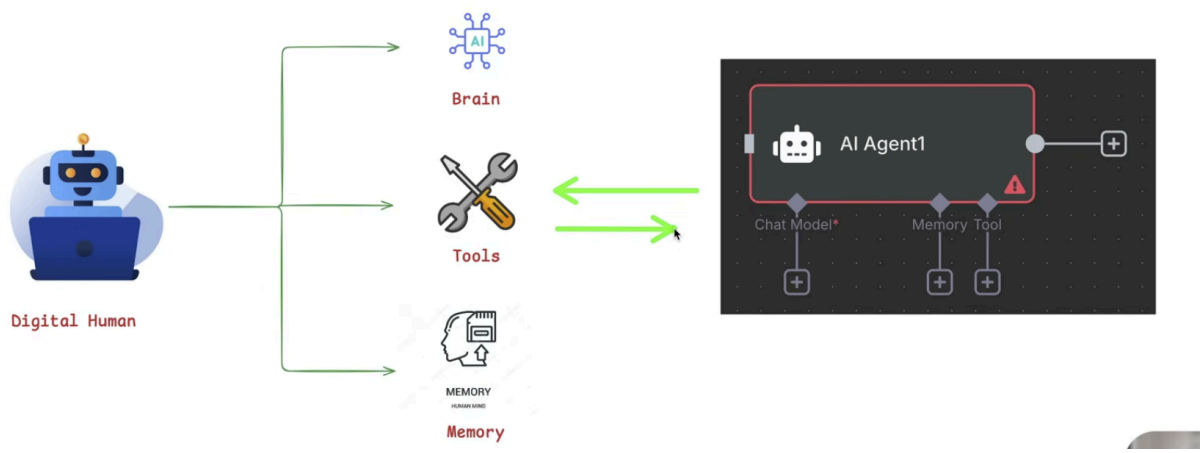
AI & Agents in n8n:



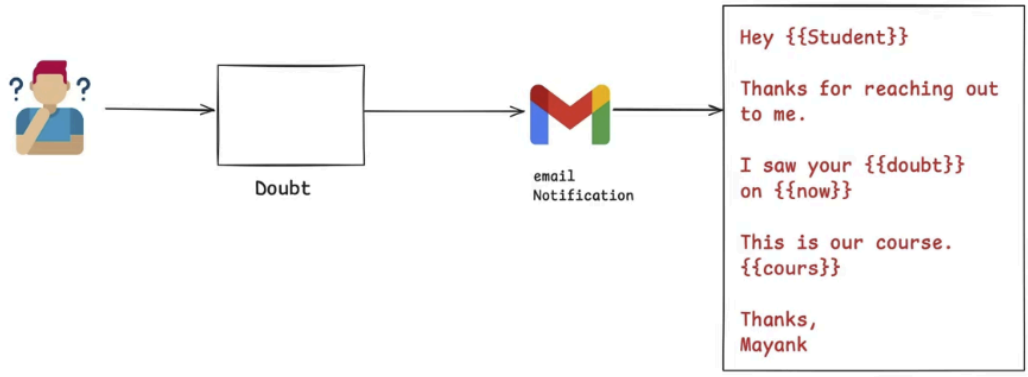
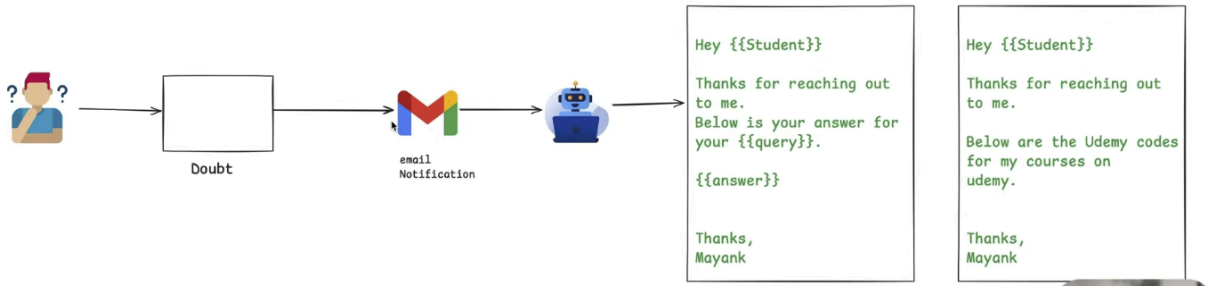


AI Agent:

Above are the AI Automation only, Now will see the AI Agent.



Automation
Rule-Based
Predefined Processes
Sequential Tasks
Limited Decision-Making
Requires Human Input to Change

AIAgent
Data-Driven
Adaptive Learning+
Complex Reasoning
Autonomous Actions
Dynamically Modifiable

Agentic AI Flow 1 - Doubt Solving Assistant